

Software Engineer

SAI SUNANDH R

Phone: +1 9285502812

Email: saisunandhramayanam@gmail.com

Professional Summary:

- **Innovative and impact-driven Software Engineer & Python Developer** with a strong focus on **ETL pipelines, data analytics, and machine learning**, especially in **healthcare, pharma, and finance** domains. I specialize in crafting **secure, scalable, and regulation-compliant data solutions** using Python, PySpark, SQL, and AWS—empowering organizations to turn complex data into actionable insights.
- I build **automated, end-to-end ETL workflows** that seamlessly integrate data from diverse sources—databases, APIs, cloud platforms—ensuring high-quality, real-time access to clean, transformed, and analytics-ready data. I optimize performance with SQL-based warehousing tools like **AWS Redshift, RDS, and MySQL** to deliver fast, reliable insights at scale.
- Skilled in building **real-time data streaming solutions**, performing **feature engineering, anomaly detection**, and applying **data augmentation techniques** to boost model performance and decision-making accuracy.
- With a passion for data storytelling, I create **interactive dashboards and visualizations** using **Tableau, Power BI, Matplotlib, Seaborn, and Plotly**, translating raw numbers into insights that drive strategy in areas like patient outcomes, vaccine logistics, and financial reporting.
- A CI/CD champion, I automate deployments using **GitHub Actions, Jenkins, and Docker**, and manage cloud-native infrastructure on **AWS (Lambda, S3, Glue, CloudWatch)** to ensure **resilient, scalable, and cost-efficient data operations**.
- I'm deeply committed to **data privacy and compliance**, implementing **IAM, encryption**, and governance best practices aligned with **HIPAA, GDPR**, and industry regulations.
- Known for being a **collaborative force** across business, engineering, and data science teams, I bring a proactive mindset, a love for learning, and a dedication to leveraging technology for smarter, faster, and more secure data solutions.

Technical Skills

Skillset	Skills
Programming Languages	Python (3.12), C, C++, Java
Machine Learning & Deep Learning	Scikit-learn, PyTorch, OpenCV
Generative AI & LLMs	GPT, BERT, LLaMA, T5, Diffusion Models
Cloud Computing	Google Cloud Platform (GCP), Amazon Web Services (AWS), Microsoft Azure
Data Engineering & ETL	Apache Spark, Kafka, Airflow, Pandas, NumPy, SciPy
DevOps & Automation	Docker, Kubernetes, Terraform, CI/CD (Cloud Build, GitHub Actions, Jenkins)
Database Management	SQL, MySQL, PostgreSQL, MongoDB
Statistical Analysis & Data Science	Data wrangling, Hypothesis testing, Feature engineering, Time-series forecasting
API Development & Backend	Flask, FastAPI, Django, RESTful APIs, OAuth2, JWT authentication
Visualization & Reporting	Matplotlib, Seaborn, Plotly, Tkinter, Google Data Studio, R's Shiny
Security & System Monitoring	IAM roles, Access control, GCP Cloud Monitoring, Logging tools
Agile Development & Collaboration	Agile methodologies, Jira, Confluence, Git
Emerging AI Trends & Innovation	Federated learning, AutoML, Reinforcement learning, Responsible AI

Certifications

- AWS Certified Data Engineer - Associate: Demonstrated expertise in architecting, deploying, and managing scalable data solutions using AWS services like S3, Redshift, Glue, and RDS.
- Microsoft Power Platform Fundamentals: Proficient in designing business solutions and workflows using Power Apps, Power BI, and Power Automate for actionable insights and process automation.
- DevOps Specialist: Advanced proficiency in CI/CD pipeline creation using Jenkins, and Infrastructure as Code (IaC) practices to ensure seamless application deployment and scalability.
- Python Data Science & Machine Learning Certification: Deep expertise in Python for data manipulation, statistical analysis, and machine learning, including libraries like Scikit-learn, Pandas, and NumPy.
- PySpark Specialist Certification: Advanced skills in distributed data processing, building scalable ETL pipelines, and executing large-scale data transformations using PySpark.
- Machine Learning & AI Specialist: Proven ability in designing, training, and deploying machine learning models for classification, regression, and clustering, with experience in hyperparameter tuning and evaluation.
- SQL Certification: Mastery in querying, managing, and analyzing structured data in relational databases, ensuring high performance and data integrity.
- JupyterLab Data Analytics & Workflow Automation Certification (Self-taught): Expert in using JupyterLab for data analysis, pipeline development, and visualizing workflows for reproducibility and collaboration.
- Robotic Process Automation (RPA) Practitioner: Skilled in automating repetitive tasks and improving operational efficiency with RPA tools.
- IoT & Data Analytics: Hands-on experience with IoT sensor integration, real-time data analysis, and designing solutions for interconnected systems.
- Web Development (Django): Expertise in developing scalable web applications using Django, with strong backend and front-end integration capabilities.

Work Experience

Python Developer Intern

Averosoft Inc

February 2025 –

Current

- Understanding of the project architecture, backend frameworks, cloud-based infrastructure, and relational/non-relational database structures. Reviewing system workflows to identify optimization and automation opportunities.
- Implementing security best practices, performance tuning, and scalable software design patterns in ongoing development efforts. Actively applying industry standards in Django, and RESTful API development to deliver clean and maintainable codebases.
- Working hands-on with CI/CD pipelines, containerization tools (Docker), and deployment strategies to support robust, fault-tolerant applications. Collaborating with DevOps teams to streamline cloud integration (AWS) and automate infrastructure components.
- Participating in agile development workflows, including sprint planning, daily stand-ups, and code reviews. Translating complex business requirements into scalable and efficient technical solutions.
- Analyzing and refactoring legacy systems by reviewing historical codebases, architecture diagrams, and API documentation to support seamless system integration and feature enhancements.
- Designing, developing, and maintaining Python-based web applications using Django . Building and consuming REST APIs, writing backend logic, and integrating with SQL/NoSQL databases.
- Writing optimized code for backend implementations, managing code refactoring, bug fixes, and performance optimizations. Conducting unit and integration testing to ensure code reliability.

- Developing and maintaining ETL pipelines, with full ownership of data cleaning, transformation, and loading processes using PySpark, Pandas, and NumPy. Ensuring data accuracy, consistency, and high performance across large-scale data systems.
- Building interactive dashboards and data visualizations in Tableau to convey insights from processed data. Collaborating with stakeholders to define KPIs, monitor trends, and enable data-informed decision-making.
- Optimizing data storage solutions, including indexing and querying techniques, to improve performance and support analytical workloads. Ensuring alignment with data privacy, security, and compliance standards such as HIPAA and GDPR.

Software Engineer
SDVS Tech
 2025

September 2024 – February

- Get trained on project-specific tools, technologies, and frameworks (Python, Django, Flask, FastAPI). Understand coding standards, best practices, and version control workflows (Git, GitHub). Learn Agile methodologies, sprint planning, and team collaboration workflows.
- Review existing codebases to understand system architecture, workflows, and dependencies. Analyze technical documentation, API specifications, and project scope.
- Shadow senior developers to gain insights into debugging, issue resolution, and deployment processes. Work on assigned project modules, implementing Python-based features and functionalities.
- Develop, test, and maintain web applications, APIs, and backend services. Optimize code for performance, scalability, and security following project requirements.
- Collaborate with the team to integrate third-party APIs, databases (SQL/NoSQL), and data processing tools. Participated in team discussions, sprint planning, and technical reviews. Maintain detailed documentation for project modules, APIs, and system architecture. Engage in code reviews, receive feedback, and apply suggested improvements.
- Optimized SQL queries and **data warehousing solutions** (MySQL, PostgreSQL, Redshift) for faster analytics.
- Automated **code deployment and CI/CD pipelines** using **Jenkins, GitHub Actions, and Docker**. **Wrote Python scripts** for system monitoring, task scheduling, and API interactions.

Teaching Assistant
Northern Arizona University
 2024

August 2023 – May

- Mentored students in Data Structures, Large Scale Data Structures, and Principles of Programming Languages (Python), providing comprehensive tutoring and grading support for approximately 150 students per course, fostering a deep understanding of data engineering principles.
- Developed personalized tutoring sessions, significantly enhancing student comprehension and performance in complex topics such as algorithm analysis, data structure optimization, and data processing techniques.

- Streamlined grading and feedback processes for assignments and exams in Data Structures, Large Scale Data Structures, and Python courses, reducing turnaround time by 30% and ensuring timely, constructive feedback, analogous to optimizing data workflows in ETL processes.
- Utilized data engineering skills to maintain accurate grade records, employing efficient data management techniques to ensure data integrity and accessibility.
- Facilitated effective communication between students and faculty, leveraging skills in data visualization and reporting to provide clear and actionable insights into student performance and progress.
- Applied principles of data analysis and data quality management to develop robust assessment methods, ensuring fair and consistent evaluation of student work. • Implemented automated solutions for repetitive administrative tasks, akin to automating data pipelines, to improve efficiency and accuracy in grade record maintenance.
- Guided students in practical applications of data structures and algorithms, preparing them for real-world data engineering challenges and enhancing their problem-solving skills.
- Conducted workshops on advanced Python programming and data processing techniques, equipping students with essential skills for data engineering roles.
- Collaborated with faculty to design and update curriculum content, integrating the latest data engineering concepts and practices to keep the coursework relevant and challenging.

Python developer

Fresenius Kabi (Accenture) - India

February

2021 – December 2022

- Utilized **Apache Spark** for distributed processing of large-scale datasets including patient records, vaccine distributions, and medical equipment logs, ensuring scalability and real-time performance.
- Implemented **event-driven and streaming data pipelines** to enable near real-time monitoring of patient health metrics and vaccine usage, improving responsiveness and proactive decision-making.
- Ensured **data integrity, traceability, and compliance** with healthcare data standards such as **HIPAA** and **GDPR**, designing pipelines that were both resilient and maintainable.
- Played a key role in **data organization and accessibility**, streamlining data flow between teams and enabling cross-functional collaboration on insights from diverse data sources like patient outcomes and inventory logistics.
- Collaborated closely with **data scientists and AI engineers** to clean, transform, and **feature-engineer datasets** using **Pandas**, **NumPy**, and **PyArrow**, supporting predictive modeling efforts for clinical and operational use cases.
- Designed and developed **interactive Tableau dashboards** to visualize trends in patient data, vaccine efficacy, and equipment usage, enabling stakeholders to make data-driven decisions with clarity and speed.
- Defined key **KPIs** and worked with business units to generate insights through **visual storytelling** in Tableau, increasing transparency and enhancing executive reporting capabilities.
- Participated in **Agile/Scrum ceremonies**, used **Jira** for sprint task management, and employed **Git** for version control, contributing to rapid iterations and continuous integration of robust data solutions.
- Implemented **data security protocols**, including **IAM**, data **encryption**, and role-based access control, to protect sensitive patient information and enforce regulatory compliance.
- Optimized **data storage** using efficient formats such as **Parquet** and **JSON**, balancing performance, storage cost, and query speed for analytics workloads.
- Leveraged **PyArrow** to manage larger-than-memory datasets, ensuring scalable data processing in high-volume environments.
- Automated **deployment and testing** of data pipelines with **Jenkins** and **Docker**, enabling reliable and consistent rollout of new features across development, staging, and production environments.

Data Analyst

Cosmo Ferrites - India

December 2019 – January 2020

- I collaborate with stakeholders to understand their specific needs around financial data, including revenue, purchase orders, scheduled deliveries, and completed orders. With clear requirements in place, I focus on extracting data from various sources such as databases, spreadsheets, and external APIs, utilizing MySQL and Python tools like Pandas and SQL Alchemy to ensure seamless integration.
- Once the data is consolidated, I preprocess and clean it by handling missing values, detecting anomalies, and standardizing formats to maintain accuracy and consistency. By leveraging libraries like NumPy and Pandas, I perform advanced calculations and derive key performance metrics, such as monthly revenue growth, delivery efficiency, and order fulfillment rates.
- To transform data into actionable insights, I utilize visualization tools like Matplotlib and Seaborn, creating dashboards that highlight revenue trends, delivery timelines, and order statuses. These visual representations enable stakeholders to identify inefficiencies and opportunities, facilitating data-driven decision-making for improved operational performance.
- Ensuring long-term scalability, I design and implement automation workflows that streamline data retrieval, processing, and updates. By utilizing in-house open-source tools along with MySQL event schedulers and Python's schedule library, I automate data pipelines, ensuring that financial and operational data remains current and readily accessible.
- To support maintainability and future enhancements, I write reusable code modules and thoroughly document workflows, enabling smooth transitions for other teams. Through these efforts, I contribute to delivering robust, scalable, and insightful solutions that empower organizations to maximize the value of their data.

Research Associate - India

April 2018 –

October 2019

- As a Research Associate, I worked on a machine learning-driven healthcare analytics project, leveraging the tools and methodologies available at the time to develop predictive models for disease detection. Given the computational constraints and limited deep learning adoption in mainstream research, the project focused on classical machine learning techniques to derive meaningful insights from structured health data.
- I utilized Python in Jupyter Notebook, employing libraries such as Pandas and NumPy for data preprocessing, ensuring data integrity through cleaning, normalization, and transformation techniques. Given the hardware constraints, I focused on feature engineering strategies like principal component analysis (PCA) and statistical correlation methods to optimize model input without excessive computational costs. Exploratory data analysis (EDA) was conducted using Matplotlib and Seaborn to visualize trends, distributions, and potential biases within the dataset, guiding the selection of relevant features for model training.
- Machine learning models, including logistic regression, decision trees, random forests, and support vector machines (SVM), were developed using Scikit-learn. Since deep learning frameworks like PyTorch were in early adoption phases and required higher computational resources, the project emphasized optimizing traditional models through hyperparameter tuning and ensemble learning techniques to achieve robust predictive performance. Hyperparameter tuning was conducted using grid search and k-fold cross-validation to balance model complexity and generalization.
- Due to the limited availability of cloud-based machine learning services, local execution and batch processing techniques were used to train and evaluate models efficiently. Model performance was assessed using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, ensuring a comprehensive understanding of prediction reliability. Results were interpreted and documented systematically to contribute to the research community's understanding of practical, resource-efficient machine learning implementations.
- To facilitate reproducibility, I collaborated with research peers and documented the entire workflow, from data acquisition and preprocessing to model evaluation. Given the limited automation tools at the time, a structured

approach was followed to manually validate findings and ensure consistency across iterations. Visualizations and summary reports were created to effectively communicate results to the research team, guiding further improvements and potential applications in public health analytics.

- The research contributed to the growing field of machine learning in healthcare, demonstrating how classical models, when carefully tuned and optimized, could provide valuable predictive capabilities even with limited computational resources. The findings laid the groundwork for future advancements in data-driven diagnostics, influencing subsequent studies in predictive healthcare analytics.